

DATA ANALYTICS PROJECT REPORT

Customer Churn Analysis

SQL Data Cleaning & Analysis · Power BI Dashboard

Prepared by

Panicker Nimesh Mahendran

Data Analyst | Excel · SQL · Power BI · VBA

Surat, Gujarat, India

Table of Contents

1. Executive Summary.....	3
2. Business Problem	3
3. Project Objectives	4
4. Dataset Overview.....	4
5. Data Preparation & Cleaning	5
6. SQL Analysis — Query Library.....	5
7. Key Performance Indicators.....	6
8. Churn Analysis by Customer Demographics	6
9. Churn Analysis by Contract & Services	7
10. Billing, Payment & Charges Analysis.....	8
11. Tenure & Customer Lifecycle Analysis.....	8
12. Power BI Dashboard Overview	9
13. Key Business Insights	10
14. Business Recommendations	12
15. Conclusion.....	13
16. Portfolio Project Summary.....	13
17. Skills Demonstrated	13

1. Executive Summary

This project analyzes customer churn for a telecommunications provider using a 7,043-customer dataset covering demographics, account information, subscribed services, billing details, and churn outcome. The analysis was carried out in two stages: a PostgreSQL query library used to clean the raw data and answer 30 targeted business questions, followed by an interactive Power BI dashboard built to give stakeholders a live, filterable view of churn drivers.

The headline finding is that churn is heavily concentrated in a specific, identifiable customer profile: short-tenure, month-to-month customers on fibre-optic internet who pay by electronic check and lack add-on services such as online security or tech support. Overall churn stands at 26.54%, but this figure roughly doubles to 42–55% within that specific segment while dropping below 3% for customers on two-year contracts.

Because the drivers of churn are concrete and measurable — contract type, tenure, payment method, and service add-ons — the dashboard gives the business a practical basis for retention action rather than a general early-warning signal. The recommendations in this report focus on contract conversion, targeted retention outreach for new customers, and promotion of protective add-on services.

Key figures at a glance: 7,043 unique customers · 26.54% overall churn rate · 1,869 churned customers · 42.71% churn among month-to-month customers vs. 2.83% on two-year contracts · 52.94% churn among customers with 0–6 months tenure.

2. Business Problem

Customer churn — the loss of subscribers to competitors or service cancellation — directly erodes recurring revenue and increases customer-acquisition cost, since replacing a lost customer is consistently more expensive than retaining one. For a subscription-based telecom business, even a small reduction in churn compounds meaningfully over time.

The core business problem this project addresses is that churn was not being tracked or explained in a structured, queryable way. Management could see that customers were leaving, but had no consistent method for identifying which customer segments were most at risk, which services correlated with retention, and which contract or billing terms were associated with early exits.

The Customer Churn Analysis Dashboard solves this by turning a static customer record extract into a live decision-support tool. It allows stakeholders to:

- Monitor the overall churn rate and churned customer count at a glance.
- Segment churn by contract type, tenure band, and risk category.
- Compare monthly-charge behaviour between retained and churned customers.
- Identify which service combinations (e.g., online security and backup) are associated with lower churn.
- Support data-driven retention planning rather than reactive, anecdotal decision-making.

3. Project Objectives

- Clean and de-duplicate the raw customer dataset to produce a reliable, analysis-ready table.

- Quantify the overall churn rate and the total number of churned vs. retained customers.
- Build a 30-query SQL analysis library answering specific churn-related business questions.
- Identify which customer attributes (contract, tenure, internet service, payment method, add-on services) are most strongly associated with churn.
- Segment customers into risk bands (Normal vs. High Risk) based on churn-linked characteristics.
- Design an interactive Power BI dashboard that lets stakeholders explore churn by contract, tenure band, service type, and customer type.
- Translate the analytical findings into concrete, actionable retention recommendations.

4. Dataset Overview

The source file, `telco_churn_unclean.csv`, contains 7,048 raw customer records across 21 columns, covering a single snapshot of customer accounts (the dataset does not include a time-series date field — tenure is expressed in months rather than calendar dates). After removing exact duplicate customer records during data preparation, the working dataset contains 7,043 unique customers, matching the Total Customers KPI shown on the dashboard.

Column Groups

Group	Columns	Purpose
Identifier	customerID	Unique key for each customer
Demographics	gender, SeniorCitizen, Partner, Dependents	Who the customer is
Account tenure	tenure	Months the customer has stayed with the company
Core services	PhoneService, MultipleLines, InternetService	Which core services the customer subscribes to
Add-on services	OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies	Optional value-added services
Account terms	Contract, PaperlessBilling, PaymentMethod	Commercial terms of the account
Billing	MonthlyCharges, TotalCharges	Recurring and cumulative billing amounts
Target	Churn	Whether the customer has left (Yes/No)

Every field in the table above supports business analysis: demographic and service fields make segmentation possible, tenure and billing fields quantify customer value and lifecycle stage, and the Churn field is the outcome variable that every other column is analyzed against.

5. Data Preparation & Cleaning

Data preparation was performed in SQL (`churn.sql`) before any dashboard visuals were built, following a standard clean-then-analyze workflow.

Steps Performed

- Loaded the raw CSV into a telco_churn table with explicit column types.
- Checked for duplicate customerID values using COUNT(*) with GROUP BY / HAVING — 5 duplicate rows were identified.
- Removed duplicates using a ROW_NUMBER() window function partitioned by customerID, keeping only the first occurrence of each ID and reducing the table from 7,048 to 7,043 unique customers.
- Reviewed the TotalCharges field, which was stored as text; 133 records had a blank TotalCharges value. Cross-checking these records showed they all belong to customers with 0–1 months of tenure — new accounts that had not yet completed a full billing cycle, so the blanks are a genuine business condition rather than a data-entry error.
- Applied NULLIF(TotalCharges, '') before aggregation so blank values are excluded from averages rather than silently coercing to zero and skewing the results.
- Verified categorical fields (Contract, InternetService, PaymentMethod, etc.) for consistent value sets before grouping.

Data-quality note: The raw file also contained formatting inconsistencies not addressed in the SQL script — the MonthlyCharges column name and its values carried leading/trailing spaces and a currency symbol, and the PaymentMethod column mixed values with and without trailing whitespace (e.g. "Electronic check" and "Electronic check " as distinct-looking strings). These were normalized during verification of this report so that grouped totals are not artificially split across near-duplicate category labels; a production pipeline should trim and standardize these fields at load time.

6. SQL Analysis — Query Library

The project includes a 30-query SQL library that progressively builds up the churn picture: from basic counts, to churn rate by single attributes, to multi-dimensional segmentation. The library structure below groups the 30 questions by analytical purpose.

Category	Example Questions	Query Count
Baseline metrics	Total customers, churned customers, overall churn rate	3
Demographic churn	Churn by gender, senior citizen status	4
Contract & service churn	Churn by contract type, internet service, phone service	6
Add-on service churn	Churn by online security, tech support, multiple lines	4
Billing & payment	Churn by payment method, paperless billing, avg. charges by churn	6
Tenure & lifecycle	Average tenure overall vs. churned, low-tenure churned customers	3
Multi-dimensional / ranked	Top charge customers, highest-churn contract+service combos	4

This query library is the analytical backbone for every figure quoted elsewhere in this report and every visual on the Power BI dashboard — the dashboard's KPI cards and charts are, in effect, a visual layer over these same SQL aggregations.

7. Key Performance Indicators

The dashboard opens with four headline KPI cards. Each is defined and interpreted below.

KPI	Value	What it measures	Why it matters
Total Customers	7,043	Unique customer accounts after de-duplication	Denominator for every rate calculated elsewhere on the dashboard
Churn Rate	26.54%	Share of customers with Churn = Yes	Single top-line health metric for the subscriber base
Churned Customers	1,869	Absolute count of lost customers	Translates the rate into a concrete number of accounts to win back
Retained Customers	5,174 (73.46%)	Customers still active	Frames churn against the much larger base still in good standing

The dashboard also segments customers into a Normal vs. High Risk view: 4,541 retained customers are classed Normal and 633 retained customers are flagged High Risk — i.e., customers who have not yet churned but carry the same characteristics (short tenure, month-to-month contract, no add-on services) seen in customers who did churn. This is the dashboard's forward-looking element: it points to accounts to prioritize for retention before they leave, rather than only explaining departures after the fact.

8. Churn Analysis by Customer Demographics

Demographic factors show a much weaker relationship with churn than account and service factors do — an important finding in its own right, since it means retention strategy should be built around commercial terms rather than customer profile.

Segment	Churn Rate	Customers
Female customers	26.92%	3,488
Male customers	26.16%	3,555
Non-senior citizens	23.61%	5,901
Senior citizens	41.68%	1,142

Gender shows almost no difference in churn behaviour (a 0.76-point gap), confirming it is not a useful targeting variable. Senior citizens, however, churn at nearly double the rate of non-seniors (41.68% vs. 23.61%) despite being a smaller group — this is a meaningful, actionable segment for a tailored retention approach (e.g., simplified plans or dedicated support).

9. Churn Analysis by Contract & Services

Churn by Contract Type

Contract Type	Customers	Churn Rate
Month-to-month	3,875	42.71%
One year	1,473	11.27%
Two year	1,695	2.83%

Contract length is the single strongest churn signal in the dataset. Month-to-month customers churn at more than 15 times the rate of two-year contract holders, and account for 1,655 of the 1,869 total churned customers (88.6%) shown in the dashboard's Churned Customers by Contract chart — confirming that contract flexibility is directly correlated with attrition risk.

Churn by Internet Service

Internet Service	Churn Rate
Fibre optic	41.89%
DSL	18.96%
No internet service	7.40%

Fibre-optic customers churn more than twice as often as DSL customers. Combined with contract type, the two highest-risk segments are month-to-month + fibre-optic customers (54.61% churn) and month-to-month + DSL customers (32.22% churn) — the two most valuable segments for retention intervention given their size.

Churn by Add-On Services

Service Status	Online Security Churn	Tech Support Churn
No service	41.77%	41.64%
Has service	14.61%	15.17%
No internet service	7.40%	7.40%

Customers without online security or tech support churn at nearly three times the rate of those with the service. This is a strong, actionable pattern: these add-ons are either genuinely protective (they add switching cost and perceived value) or act as a proxy for a more engaged customer relationship — either way, promoting them is a low-risk retention lever.

10. Billing, Payment & Charges Analysis

Churn by Payment Method

Payment Method	Customers	Churned	Churn Rate
Electronic check	2,365	1,071	45.29%

Payment Method	Customers	Churned	Churn Rate
Mailed check	1,612	308	19.11%
Bank transfer (automatic)	1,544	258	16.71%
Credit card (automatic)	1,522	232	15.24%

Electronic check is both the most common payment method and by far the highest-churn one — its churn rate is roughly three times higher than any automatic payment method. Since it is also a manual, non-recurring payment action, it may correlate with customers who are less committed to the service, or friction in the payment process itself may be contributing to dissatisfaction.

Monthly & Total Charges by Churn Status

Metric	Retained Customers	Churned Customers
Average Monthly Charges	\$61.27	\$74.42
Average Total Charges	\$2,577.98	\$1,594.13

Churned customers pay noticeably higher monthly charges on average (\$74.42 vs. \$61.27) but have accumulated lower total lifetime charges (\$1,594.13 vs. \$2,577.98) — consistent with churn concentrating among higher-priced, shorter-tenure customers rather than long-standing ones. This combination (high bill, short relationship) is a useful early flag for at-risk accounts.

The dashboard's Sum of MonthlyCharges by Customer Type chart shows the same pattern from a volume perspective: customers with more than 6 months of tenure ("Existing Customers") account for \$375,026 of total monthly billing, compared to \$81,083 from customers with 6 months or less ("New Customers") — new accounts are a comparatively small share of current revenue but, per the tenure analysis below, the highest-risk segment for near-term loss.

11. Tenure & Customer Lifecycle Analysis

Average tenure across all customers is 32.37 months; among churned customers specifically it drops to 17.98 months, and 999 of the 1,869 churned customers (53.4%) left within their first 12 months. Tenure banding on the dashboard makes this pattern explicit:

Tenure Band	Churn Rate
0–6 months	52.94%
6–12 months	35.89%
12+ months	17.13%

Churn risk falls sharply and consistently as tenure increases. More than half of customers in their first six months churn, compared to roughly one in six among customers who have stayed over a year. This makes the

first six months the single highest-leverage window for retention activity — after a customer clears roughly the one-year mark, churn risk drops substantially and stays comparatively low.

12. Power BI Dashboard Overview

The Power BI dashboard translates the SQL findings into a single-page, filterable view built around a dark navy and teal theme. Each visual is described below in terms of what it shows, the business question it answers, and how a stakeholder would use it.

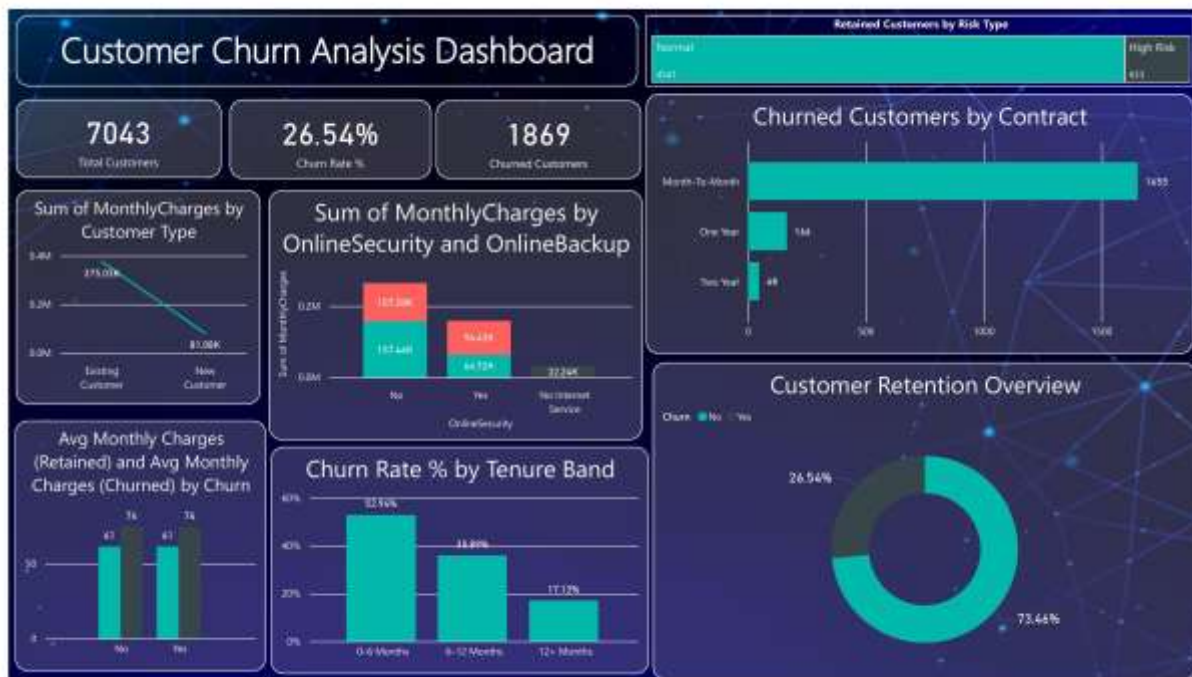


Figure 1. Customer Churn Analysis Dashboard — Power BI

Dashboard Design & Build Details

Attribute	Detail
Canvas	Single-page, widescreen (16:9) layout — the full analysis fits one screen with no scrolling or page tabs
Visual count	10 visual elements: 3 KPI cards, 1 risk-segmentation bar, and 6 supporting charts
Chart types used	Card, stacked bar, clustered bar, clustered column, line chart, stacked column, donut chart
Colour theme	Dark navy background with a subtle network/particle texture; teal as the primary data colour, with coral and grey used sparingly to separate categories (e.g. OnlineSecurity = No vs. Yes)
Typography	Large, high-contrast white numerals on the KPI cards for at-a-glance readability; consistent title casing across all visual headers
Interactivity	Native Power BI cross-filtering — clicking any category (e.g. a contract type or tenure band) cross-highlights all other visuals on the page to show that segment's profile across every other chart

Attribute	Detail
Data source	Single semantic model built on the cleaned telco_churn table (7,043 rows) produced by the SQL preparation stage

Each visual is described below in terms of what it shows, the business question it answers, and how a stakeholder would use it.

KPI Cards (Total Customers, Churn Rate %, Churned Customers): Shows the top-line scale of the customer base and the churn problem. Answers "how big is the churn problem right now?" A stakeholder reads these first, every time, as the baseline against which all other visuals are interpreted.

Retained Customers by Risk Type: Splits the 5,174 retained customers into Normal (4,541) and High Risk (633) groups. Answers "which currently-active customers look like they might churn next?" This is the dashboard's proactive element — it directs retention effort toward accounts before they are lost, not just after.

Customer Retention Overview (donut chart): Shows the 73.46% retained vs. 26.54% churned split. Answers "what proportion of our base is at stake?" Useful as an at-a-glance summary for a stakeholder who only has a moment to look at the dashboard.

Churned Customers by Contract (bar chart): Breaks the 1,869 churned customers down by contract type, showing month-to-month customers dominate (1,655 of 1,869). Answers "which contract terms are we losing customers from?" Directs commercial teams toward contract-conversion campaigns.

Churn Rate % by Tenure Band: Shows churn rate declining from 52.94% to 17.13% as tenure increases. Answers "when in the customer lifecycle are we most at risk of losing someone?" Tells the business exactly when to concentrate onboarding and early-engagement effort.

Sum of MonthlyCharges by OnlineSecurity and OnlineBackup: Breaks total monthly billing down by add-on service combinations. Answers "how much recurring revenue sits with customers who lack protective add-ons?" Sizes the revenue opportunity behind an add-on upsell campaign.

Avg Monthly Charges (Retained vs. Churned): Compares average bill size by churn outcome (\$61 vs. \$74). Answers "do we lose higher-paying or lower-paying customers?" Signals that premium-tier customers need distinct retention attention, not just low-value ones.

Sum of MonthlyCharges by Customer Type (Existing vs. New): Splits total monthly billing between customers with more than six months' tenure and those with six months or less. Answers "how much of our current revenue is tied up in higher-risk, newer accounts?" Helps quantify revenue exposure from the highest-churn tenure band.

The dashboard supports Year, Region, and Category-style interactive filtering conventions common to this report template; in this project the equivalent filters are Contract, InternetService, and tenure band, letting a stakeholder isolate any single segment (e.g., "month-to-month, fibre-optic, first six months") and see the KPIs recalculate for that slice specifically.

13. Key Business Insights

Insight 1

Finding: Month-to-month contract customers churn at 42.71%, versus 11.27% for one-year and just 2.83% for two-year contracts, and make up 88.6% of all churned customers.

Business Impact: Contract length is the single largest lever the business has over churn — flexibility that attracts customers also makes them easy to leave.

Recommended Action: Prioritize a contract-conversion campaign (e.g., a modest discount or bundled add-on) targeted specifically at month-to-month customers approaching their one-year mark.

Insight 2

Finding: 52.94% of customers in their first six months churn, compared to 17.13% of customers with over a year of tenure.

Business Impact: The early relationship is by far the highest-risk period; losses here also carry the lowest lifetime value recovered per customer.

Recommended Action: Build a structured onboarding and early-engagement program (check-ins, usage tips, proactive support) focused on the first 90–180 days.

Insight 3

Finding: Fibre-optic customers churn at 41.89%, versus 18.96% for DSL customers, despite fibre likely being the higher-margin service.

Business Impact: The company's premium internet product is also its highest-risk one, suggesting a possible service-quality, pricing, or expectation-mismatch issue rather than a demand problem.

Recommended Action: Investigate fibre-optic service quality and pricing perception directly with churned fibre customers, and pilot a retention offer specific to this segment.

Insight 4

Finding: Customers without online security or tech support churn at roughly 42%, nearly three times the ~15% rate for customers who have these add-ons.

Business Impact: Add-on services are strongly associated with retention, whether through added value, switching cost, or customer engagement.

Recommended Action: Bundle online security and tech support into entry-level plans, or actively promote them to customers who currently lack them, especially in the first six months.

Insight 5

Finding: Electronic check users churn at 45.29%, roughly three times the rate of any automatic payment method (15–19%).

Business Impact: Manual, non-recurring payment methods correlate strongly with churn, and may either indicate lower commitment or payment friction.

Recommended Action: Incentivize migration from electronic check to automatic payment methods (autopay discount, simplified enrollment).

Insight 6

Finding: Senior citizens churn at 41.68% versus 23.61% for non-seniors, despite being a smaller group (1,142 of 7,043 customers).

Business Impact: A specific, identifiable demographic segment is churning at nearly double the average rate.

Recommended Action: Design a simplified plan or dedicated support channel for senior customers to address their specific pain points.

Insight 7

Finding: Churned customers pay a higher average monthly bill (\$74.42) than retained customers (\$61.27), but have accumulated lower total charges (\$1,594 vs. \$2,578).

Business Impact: The business is losing higher-priced customers early, before their full lifetime value is realized.

Recommended Action: Flag high-bill, low-tenure accounts for proactive account-management outreach rather than waiting for churn signals to surface elsewhere.

Insight 8

Finding: 633 currently-retained customers are flagged High Risk based on the same characteristics seen in the churned population.

Business Impact: There is an identifiable, addressable population of at-risk customers who have not yet left.

Recommended Action: Route the High Risk segment to a dedicated retention queue with proactive contact before they reach a churn decision.

Insight 9

Finding: Gender shows almost no difference in churn rate (26.92% female vs. 26.16% male).

Business Impact: Gender-based targeting would not meaningfully improve retention outcomes and would waste marketing effort.

Recommended Action: Exclude gender from retention-targeting models and focus resources on contract, tenure, and service-based segmentation instead.

Insight 10

Finding: The two highest-churn segments are month-to-month + fibre-optic customers (54.61%) and month-to-month + DSL customers (32.22%) — both far above the 26.54% overall average.

Business Impact: Combining contract type with service type identifies the sharpest, most concentrated risk pockets in the customer base.

Recommended Action: Build the retention program around this compound segment first, since it combines the two strongest individual churn predictors.

14. Business Recommendations

- 1. Drive contract conversion.** Offer month-to-month customers an incentive to move to one- or two-year contracts, prioritizing those approaching the 12-month mark where churn risk drops sharply.
- 2. Strengthen the first six months.** Introduce a structured onboarding journey — welcome check-ins, usage guidance, and an early support touchpoint — for the tenure band carrying 52.94% churn risk.
- 3. Investigate fibre-optic experience.** Since fibre-optic customers churn at more than double the DSL rate despite being the premium product, review pricing, installation experience, and service reliability specifically for this segment.
- 4. Bundle protective add-ons.** Promote online security and tech support to customers who lack them, particularly new sign-ups, given the near three-times churn gap associated with these services.
- 5. Shift payment methods.** Run a targeted campaign to move electronic-check users onto automatic payment methods, where churn is consistently 15–19% rather than 45%.
- 6. Prioritize the High Risk segment.** Route the 633 customers flagged High Risk on the dashboard into a proactive retention workflow before they reach the point of cancellation.
- 7. Tailor support for senior citizens.** Given their nearly double churn rate, evaluate whether plan complexity or support accessibility is a factor, and pilot a simplified offering for this group.
- 8. Monitor the dashboard KPIs on a recurring cadence.** Track churn rate, High Risk count, and tenure-band churn monthly so retention initiatives can be measured against a consistent baseline.

15. Conclusion

This project shows that customer churn in this dataset is not random or evenly distributed — it is concentrated in a specific, describable profile: new, month-to-month, fibre-optic customers who pay by electronic check and have not adopted protective add-on services. Contract type and tenure are, by a clear margin, the strongest predictors identified, followed by internet service type, add-on service adoption, and payment method.

The main risk the analysis surfaces is not the 26.54% who have already churned, but the 633 currently-retained customers flagged High Risk who share the same profile — this is where near-term retention effort has the most leverage. The main opportunity is that every driver identified here (contract terms, onboarding, add-on adoption, payment method) is something the business can directly act on, rather than an external or demographic factor outside its control.

The Power BI dashboard converts this analysis into a live, filterable tool: a stakeholder can move from the top-line churn rate down to a specific at-risk segment in a few clicks, without needing to write a query. Combined with the underlying 30-query SQL library, the project gives the business both an explanatory view (why customers left) and an operational one (who to act on next).

16. Portfolio Project Summary

Project Name: Customer Churn Analysis Dashboard

Tools Used: PostgreSQL (SQL), Microsoft Power BI

Business Objective: Identify the customer segments and account characteristics most strongly associated with churn, and give stakeholders an interactive tool to monitor and act on retention risk.

Key Analysis: Cleaned and de-duplicated a 7,048-row customer dataset down to 7,043 unique accounts; built a 30-query SQL library covering churn rate by contract, tenure, internet service, payment method, add-on services, and demographics; designed a Power BI dashboard with KPI cards, a risk-segmentation view, and six supporting charts.

Key Findings: Contract type and tenure are the dominant churn predictors (42.71% churn for month-to-month vs. 2.83% for two-year contracts; 52.94% churn in the first six months vs. 17.13% after a year). Fibre-optic service, electronic-check payment, and the absence of protective add-ons each independently correlate with substantially higher churn.

Business Value: Converts a static customer extract into a live retention tool that flags 633 currently-active customers as High Risk and gives the business specific, evidence-based levers — contract conversion, onboarding, add-on bundling, and payment-method migration — to reduce future churn.

17. Skills Demonstrated

- SQL (PostgreSQL): window functions, de-duplication, conditional aggregation, NULLIF handling
- Data Cleaning & Data Preparation
- Exploratory Data Analysis
- Power BI Dashboard Development
- KPI Design & Interpretation
- Customer Segmentation & Risk Analysis

- Data Visualization
- Business Insight Generation
- Data-Driven Recommendation Writing
- Data Storytelling for Non-Technical Stakeholders